1. Calculate the total resistance in ohms

(a) series

$R_s = 20+30+50$

$= 100\,\Omega$

(b) parallel.

$\frac{1}{R_p} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$

$= \frac{1}{0.08}$

$= 12.5\,\Omega$

2. Calculate the total current from power supply

series

$I = \frac{V}{R}$

$= \frac{125}{100}$

$= 1.25\ A$

~~parallel~~

parrallel

$I = \frac{125}{12.5}$

$= 10A$

3. Calculate current through each resistor.

series $= 1.25A$

parallel

$R_1 = \frac{125}{20}$

$= 6.25A$

$R_2 = \frac{125}{100}$

$= 1.25A$

$R_3 = \frac{125}{50}$

$= 2.5A$

(4) Calculate voltage drop through each resistor

series

$V = 1.25 \times 20$

$\boxed{= 20V}$

$V = 1.25 \times 30$

$\boxed{= 37.5V}$

$V = 1.25 \times 50$

$\boxed{= 62.5V}$

parrallel

$\boxed{V = 125V}$

5. Calculate the power dissipated through each resistor.

Power = Voltage * Current = VI

Series

P = VI

P = 25 x 1.25

= 31.25 W

P = 37.5 x 1.25

= 46.875 W.

P = 62.5 x 1.25

= 78.125 W.

Parallel

P = VI

P = 125 x 6.25

= 781.25 W

P = 125 x 1.25

= 156.25 W

P = 125 x 2.5

= 312.5 W

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